The Significance of Accessible Volume

A way to observe the shape of a model's next decision
before it happens

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Abstract

A language model's output tells us what it generated. It does not, by itself, tell us how many plausible continuations preceded that output. Accessible volume addresses this gap. The metric counts the next tokens whose probabilities exceed a chosen threshold. It turns a high-dimensional probability distribution into a direct measurement of how open or concentrated the next-token decision is. This paper explains the significance of that measurement and demonstrates it with a real Ablation + GEM experiment. In the measured context, only four tokens exceed probability 0.02, two exceed 0.05, and one exceeds 0.15. The result shows that the model's immediate decision surface is concentrated on a small set of alternatives. The significance is practical: accessible volume makes the shape of an impending choice observable before the choice is made.

1 The missing information in an output

A generated answer is a record of what happened. It is not a complete description of the decision that produced it.

Two model states can produce the same next token while having very different distributions of alternatives. In one state, the selected token may dominate every other possibility. In another, many tokens may remain plausible. Reading only the final token hides that distinction.

The relevant question is therefore not merely, "What did the model choose?" It is also, "How much room did the model have to choose differently?" Accessible volume provides one way to answer that question with a number.

2 Origin of the research

This work builds on Nicholas Michael Kloster's March 2026 paper, *Ambiguity, Autonomy, and the Autoregressive Compliance Cascade*. That earlier security-research finding documented a retrospective discovery through interaction with Anthropic's Claude: ambiguity could establish a favorable context, and later generation could reinforce it through an autoregressive compliance cascade. It introduced Ambiguity Front-Loading (AFL) as the observed case study and motivated direct examination of output distributions.

The discovery account identifies **March 4, 2026** as the date of first observation. It was not a deliberately engineered penetration test of the model. The pattern was recognized retrospectively in an otherwise routine research conversation: an ambiguous beginning established a favorable frame, subsequent turns accumulated actions within that frame, and the original context was not independently re-evaluated. The companion disclosure records the discovery circumstances and later reproduction.

The original case study and the present experiment are different forms of evidence. The earlier paper reports one production conversation and proposes a trajectory-level interpretation. The present work performs a separate, controlled next-token measurement using actual Ablation + GEM evidence. It does not independently reproduce AFL or claim that a single-step count detects a cascade. Both original documents are embedded in this PDF as part of the research record.

3 The equation

Let S be the full input context and $\mathcal{O}$ a finite output vocabulary. Let $P(a \mid h_T(S))$ be the probability that the model assigns to token a as the next token after processing S. For a threshold τ, accessible volume is

$$A_\tau(S) = \sum_{a \in \mathcal{O}} \mathbf{1}[P(a \mid h_T(S)) > \tau]\,. \tag{1}$$

The indicator is one when the probability is strictly greater than τ and zero otherwise. Tokens exactly at the threshold are excluded. The count measures the number of continuations that remain above the selected probability bar.

A normalized form divides the count by vocabulary size:

$$A_{\mathrm{norm},\tau}(S) = \frac{A_\tau(S)}{|\mathcal{O}|}. \tag{2}$$

This reports the fraction of the vocabulary above the threshold. It does not make different models directly comparable.

The computation is simple: after obtaining the distribution, evaluate one comparison per token. The important part is not computational novelty. It is the decision to observe the distribution as an object of measurement rather than treating only the generated text as evidence.

4 Why the measurement matters

4.1 It distinguishes a forecast from an outcome

The probability distribution is the forecast. Sampling or taking the maximum selects an outcome. Accessible volume describes the shape of the forecast without committing to that outcome.

A small count indicates that the model has concentrated its above-threshold alternatives. A large count indicates a broader set. The measurement does not say which token will win. It says how much of the vocabulary remains meaningfully in play at the selected threshold.

4.2 It makes concentration observable

Concentration is usually discussed through entropy, maximum probability, or top-token mass. Accessible volume adds a different description: the size of the region above a chosen cutoff.

The three descriptions are not interchangeable. A distribution can have one dominant token and few above-threshold alternatives, or one dominant token and many alternatives just below the threshold. The count captures a boundary that a single maximum or entropy value does not fully express.

The threshold sweep is more informative than one reading. As τ increases, $A_\tau(S)$ cannot increase. Watching where alternatives disappear reveals whether probability is concentrated in a narrow elite or spread across a broad collection of viable continuations.

4.3 It creates a boundary for inspection

Ablation + GEM is built around the interaction of a stochastic agent, deterministic execution, shared memory, and human judgment. The agent proposes. The environment executes and records. The human directs and reviews.

Accessible volume supplies a way to inspect the agent before its proposal becomes an action. Actual investigation evidence can be placed into a model context, and the resulting distribution can be measured without generating another token. The deterministic environment can then preserve that measurement alongside the rest of the evidence.

This does not make the metric an authorization mechanism. It makes it an instrument for understanding the agent's immediate decision surface while the rest of the system remains constrained and inspectable.

5 Evidence from Ablation + GEM

The measurement was performed on a real forward pass using 963 tokens of actual Ablation + GEM investigation context. The context included a selected artifact, a scoped observation, and real entropy-inspection output from Ablation. A local autoregressive model returned a full distribution over 50,257 next tokens.

Table 1: Measured accessible volume for the actual Ablation + GEM context.

Threshold τ	Accessible volume $A_\tau(S)$
0.02	4
0.05	2
0.10	2
0.15	1
0.30	1
0.50	1

The distribution is concentrated. A newline has probability 0.535587. End-of-text follows with probability 0.109893. The ten most probable tokens contain 75.45% of the total probability mass. Entropy over the full distribution is 4.2998 bits.

The immediate meaning is straightforward: the model is not spreading its above-threshold probability across many alternatives. Its next-token decision is concentrated on a small set. Accessible volume turns that observation into a precise, reproducible statement rather than an impression based on the final text.

A repeated forward pass on the same context produced identical input IDs and probability values. Controlled context perturbations changed some threshold counts and left others unchanged. This confirms that the instrument measures the distribution associated with a particular model and context rather than a universal property of the prompt.

6 What the result does not establish

The experiment does not establish that a concentrated choice is a correct choice. It does not establish that a broad choice is a better choice. It does not demonstrate how later tokens will behave, and it does not turn a single-step measurement into a prediction of a complete response.

The measured model is also not interchangeable with Kimi, Claude, or Ablation's semantic encoders. Those systems require their own probability distributions before equivalent measure-

ments can be made. A distribution over command-text tokens is not automatically a distribution over complete structured actions.

These limits do not weaken the measurement. They define its proper use. Accessible volume describes the shape of a particular forecast at a particular moment. Broader interpretations require separate evidence.

7 Possible paths opened by the equation

The measurement also suggests directions for further work, without establishing them as completed findings.

Prediction-aware defense. Repeated observations could provide early-warning metadata for an agent firewall. An unusual change in the distribution might lead to additional review, a read-only restriction, or a request for human judgment. It would not prove that an attack had occurred. Any defensive response would still need deterministic policy and signed evidence for later investigation.

From tokens to structured actions. A future experiment could examine distributions over complete permitted actions rather than command-text tokens. This might help a system prepare for several plausible next steps while preserving the same constraints on every action. It would not identify an adversary's intent or make an action safe by virtue of being predicted.

Authenticated, evolving evidence. Prediction observations could become part of a signed evidence record whose keys rotate over time. Standard cryptographic key derivation and authenticated encryption would be required. A timestamp could determine when to rotate a key, but neither timing nor accessible volume should determine the key itself. The metric is not an entropy source or a faster route to cryptanalysis.

These are possible paths for investigation, not claims established by the present experiment.

8 Conclusion

The significance of accessible volume is that it makes a previously hidden aspect of generation visible: the breadth of the next-token decision before selection occurs. It gives Ablation + GEM a direct way to observe whether a model's immediate choice is concentrated on few alternatives or spread across many.

The measured example shows the instrument working in practice. Four tokens exceed 0.02, two exceed 0.05, and one exceeds 0.15. That is the finding: a real investigation context produced a concentrated next-token decision surface, and the equation made it measurable.

A Ablation + GEM as the working implementation

Ablation is the reverse-engineering framework. GEM (Graphics Environment Manager) is its graphical environment and shared investigation Lab. The public implementation is available at https://github.com/Ablation-Tool/ablation.

Ablation + GEM is itself a working demonstration of the chamber model. The AI Agent supplies probabilistic labor and proposals. Ablation + GEM supplies constrained execution, persistent state, verification, and replay. The human supplies direction and judgment. Investigation events, selected objects, evidence, and verification receipts form a unified memory architecture shared by the human and agent.

The implementation used for this experiment is revision d596bed3. At that revision, the main Ablation CI passed 2,803 tests with 93 skips, and native Linux, Windows, and macOS workflows passed. Those results demonstrate the implementation's behavior. The accessible-volume experiment adds a separate measurement using actual session evidence produced by that implementation; it is not a claim that the metric was already part of the framework.

B Evidence manifest

The experiment used the public distilgpt2-ONNX model and ONNX Runtime 1.31.1. Table 2 records the identities needed to verify the evidence independently.

Table 2: Evidence identity and scope.

Item	Value
Model SHA-256	0f79f192f3137e9a90213f1b5d4a52b6d896
Tokenizer SHA-256	b735f9cb82160e5de7cc6e539c41 a6aa29bf8416d74ad795a73262b1aa3f98
Model configuration SHA-256	5564ee338adbee7ffbc5861f78b6b8 dffb0253be590126da5dc43b75475f6f
Probability values SHA-256	6ef4576439dec46e8ec83bf0408fcf83 5f6313599e77e72947435e52b273054fab
Input IDs SHA-256	da1e97006d0636658350ac253ed638 970665aaaafdd2964012f3759b993e53
Entropy evidence SHA-256	dfe43e6d1793edc68a6b3896089fdbf8 fd551e583e54753744900d66acee85b9
Original AFL paper SHA-256	9a6c07bdcef123f4fd92186a0cc8880d dec64904f5954e29b44105ba6d41a97f38915ffbecf972554e4eda70d8abbebc
AFL disclosure SHA-256	bfa75181d631aca4f8a95d0344819ec6c80ad3ac16ac9cd92ce3986e03055196
Context length	963 tokens
Vocabulary size	50,257 tokens
Probability sum	Approximately 1.0 within floating-point precision

The raw NPZ artifact contains the complete input-ID sequence and every probability value, not only the ten largest values. The JSON report contains the exact prompt, model metadata, threshold sweep, top tokens, and scope. The full entropy output and model configuration are reproduced below. The context-sweep artifact contains every controlled prompt variant and its raw probability vector.

C Reproduction procedure

The following procedure reproduces the numerical measurement in the tested CPU environment. It uses the public Ablation repository and the public ONNX model, with no API credentials.

1. Clone https://github.com/Ablation-Tool/ablation and check out revision d596bed31ba5fe361efa1c6fdad6d86e3126ab63.
2. Create an isolated Python 3.14 environment and install Ablation, GEM Lab, NumPy 2.3.5, ONNX Runtime 1.31.1, Tokenizers 0.23.3, and Hugging Face Hub 1.33.0.
3. Obtain the measurement harness embedded in this PDF and place it in a writable working directory.
4. Run `python ablation-gem-measure.py -project /path/to/ablation -output results`. The harness downloads and hash-checks the model, constructs real Lab evidence, runs inference, computes the sweep, and reloads the resulting probability vector for verification.
5. Confirm the reference sweep $5, 4, 3, 2, 1, 0$ and the actual sweep $4, 2, 2, 1, 1, 1$ at thresholds $0.02, 0.05, 0.10, 0.15, 0.30, 0.50$.

6. Load the embedded probability vector and independently recompute each count using the strict comparison $P(a \mid h_T(S)) > \tau$. Compare the raw-value and input-ID hashes with the manifest.

The numerical values and raw-value hashes are the reproducible evidence. Timing values and archive-container metadata may differ between runs.

D Controlled context evidence

The baseline was repeated before perturbing the context. It produced identical input IDs and probability values. Table 3 reports the controlled follow-up measurements.

Table 3: Accessible volume across controlled context variants.

Variant	Tokens	0.02	0.05	0.10	0.15	0.30	0.50
Baseline	963	4	2	2	1	1	1
Metadata only	522	4	2	2	2	1	0
Abbreviated evidence	295	4	2	2	2	1	1
No evidence	103	3	1	1	1	1	1
Summary request	968	3	2	1	1	1	0
Repeated short evidence	487	3	2	2	1	1	1

These differences are observations about specific model/context pairs. They do not establish a universal rule about which context increases or decreases breadth.

E Complete Ablation entropy evidence

```
BinaryContext: true
  sha256     : 913a39cd38f35349...
  arch       : x86_64
  base_va    : 0x0
  func_starts: 20
  exports    : 2
  plt entries: 33
  strings    : 65
  call_edges : 136
  str_xrefs  : 89 pairs indexed
  named funcs: 0 (overlay)

  exports (first 2):
    0x257f  error
    0x26c8  error_at_line

  imports via PLT (first 20):
    0x2250  __vfprintf_chk
    0x2260  abort
    0x2270  __errno_location
    0x2280  strncmp
    0x2290  _exit
    0x22a0  __fpending
    0x22b0  fcntl
    0x22c0  textdomain
    0x22d0  fclose
    0x22e0  bindtextdomain
    0x22f0  dcgettext
    0x2300  __ctype_get_mb_cur_max
    0x2310  strlen
    0x2320  __stack_chk_fail
    0x2330  __overflow
    0x2340  strrchr
    0x2350  mbrtoc32
    0x2360  memcmp
```

```
    0x2370  fputs_unlocked
    0x2380  strcmp

Entropy regions (26 found):
  0x00000000 - 0x00000400  (     1,024 B)  H=2.291  SPARSE
  0x00000400 - 0x00000700  (       768 B)  H=0.801  PADDING
  0x00000700 - 0x00000a00  (       768 B)  H=3.128  SPARSE
  0x00000a00 - 0x00000b00  (       256 B)  H=4.586  CODE
  0x00000b00 - 0x00000d00  (       512 B)  H=2.972  SPARSE
  0x00000d00 - 0x00002000  (     4,864 B)  H=0.271  PADDING
  0x00002000 - 0x00002400  (     1,024 B)  H=3.973  SPARSE
  0x00002400 - 0x00005300  (    12,032 B)  H=5.184  CODE
  0x00005300 - 0x00005400  (       256 B)  H=4.333  SPARSE
  0x00005400 - 0x00006000  (     3,072 B)  H=0.000  PADDING
  0x00006000 - 0x00006300  (       768 B)  H=4.865  CODE
  0x00006300 - 0x00006400  (       256 B)  H=3.591  SPARSE
  0x00006400 - 0x00006500  (       256 B)  H=4.503  CODE
  0x00006500 - 0x00006600  (       256 B)  H=4.477  SPARSE
  0x00006600 - 0x00006700  (       256 B)  H=4.980  CODE
  0x00006700 - 0x00006c00  (     1,280 B)  H=3.535  SPARSE
  0x00006c00 - 0x00006d00  (       256 B)  H=4.970  CODE
  0x00006d00 - 0x00006f00  (       512 B)  H=3.247  SPARSE
  0x00006f00 - 0x00007d00  (     3,584 B)  H=0.060  PADDING
  0x00007d00 - 0x00007e00  (       256 B)  H=1.535  SPARSE
```

F Model configuration

```
{
  "_name_or_path": "distilbert/distilgpt2",
  "_num_labels": 1,
  "activation_function": "gelu_new",
  "architectures": [
    "GPT2LMHeadModel"
  ],
  "attn_pdrop": 0.1,
  "bos_token_id": 50256,
  "embd_pdrop": 0.1,
  "eos_token_id": 50256,
  "id2label": {
    "0": "LABEL_0"
  },
  "initializer_range": 0.02,
  "label2id": {
    "LABEL_0": 0
  },
  "layer_norm_epsilon": 1e-05,
  "model_type": "gpt2",
  "n_ctx": 1024,
  "n_embd": 768,
  "n_head": 12,
  "n_inner": null,
  "n_layer": 6,
  "n_positions": 1024,
  "reorder_and_upcast_attn": false,
  "resid_pdrop": 0.1,
  "scale_attn_by_inverse_layer_idx": false,
  "scale_attn_weights": true,
  "summary_activation": null,
  "summary_first_dropout": 0.1,
  "summary_proj_to_labels": true,
  "summary_type": "cls_index",
  "summary_use_proj": true,
  "task_specific_params": {
    "text-generation": {
      "do_sample": true,
      "max_length": 50
    }
  },
  "transformers_version": "4.43.4",
  "use_cache": true,
  "vocab_size": 50257
}
```

G Embedded evidence

The complete evidence package is embedded in this PDF as attachments. It includes the exact prompt and report, all 50,257 probabilities, all 963 input IDs, the full entropy output, the model configuration, both measurement harnesses, the artifact writer, and every controlled context variant with its raw probability vector. No external filesystem is needed to inspect the evidence.

H Source signature

Signatory: Nicholas Michael Kloster [1505393089]

Timestamp (UTC): 2026-11-03T20:07:55Z

The UTF-8 LaTeX source payload preceding this section has SHA-256:

`d453ad69d807d334abd1fd4805bfeee99c3d9beea1b049e1bd1731c8825ef5d2`

This digest binds the named signatory to the exact source payload. It is an integrity record, not a third-party identity certificate.